

Brazilian E-Commerce Business Performance Analysis

Domain: E-Commerce Operations & Customer Analytics

Dataset: Brazilian E-Commerce Public Dataset (Olist)

Tools: Python (Pandas, NumPy), Seaborn, Matplotlib

Executive Summary

This project analyzes the business and operational performance of a Brazilian e-commerce marketplace across four key dimensions: revenue generation, delivery logistics, customer loyalty, and payment behavior. The objective is to identify revenue drivers, operational bottlenecks, and customer retention risks using multi-dataset aggregation and business KPIs.

The analysis shows that revenue is highly concentrated across a few product categories, sellers, and regions, particularly São Paulo. Delivery delays remain a critical operational issue and have a strong negative impact on customer ratings and repeat purchasing. Customer retention is weak, with the majority of customers making only a single purchase despite repeat buyers contributing significantly higher value. Payment behavior indicates that installment payments increase order value, while non-instant payment methods are associated with higher delivery delays.

Overall, the findings highlight that long-term growth is constrained less by customer acquisition and more by delivery reliability, seller performance, and retention-focused strategy.

Data Overview

The analysis uses transactional, customer, seller, payment, and review data from the Brazilian E-Commerce dataset. Metrics were calculated at appropriate analytical levels (order-level, customer-level, seller-level) after cleaning, joining, and feature engineering.

Key engineered features include order value, delivery time, late delivery indicators, repeat customer flags, and installment payment indicators.

Methodology

- Data cleaning and preprocessing across multiple tables
- Feature engineering for delivery time, lateness, AOV, and repeat behavior

- Aggregation at correct business levels (orders, customers, sellers)
 - Exploratory and comparative analysis using visualizations
 - Business interpretation based on observed patterns and KPIs
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1. Sales & Revenue Performance

Business Objective: Understand how money is generated.

Key Findings

Top Revenue-Generating Categories

Health & Beauty, Watches & Gifts, and Bed Bath & Table are the highest revenue-generating categories, each contributing over 1 million in sales. These categories act as core revenue drivers and should be prioritized for marketing, inventory planning, and delivery optimization.

Geographic Revenue Concentration

Sales are highly concentrated in São Paulo (SP), indicating uneven geographic demand. While SP represents the strongest revenue market, this concentration also presents risk. Premium delivery and retention strategies should focus on SP, while targeted expansion strategies can be used to grow underperforming regions.

Monthly Sales Trends

Revenue shows clear seasonality. November records the highest average monthly sales, while September has the lowest. This suggests that inventory planning and marketing spend should peak ahead of November, while promotions during September could help stabilize demand.

Average Order Value (AOV) by Category

The Computers category has the highest AOV, making it the most valuable per transaction. Focusing premium services and targeted promotions on high-AOV categories can meaningfully improve overall revenue.

Seller Revenue Contribution

Revenue is highly concentrated among a small group of sellers. Strengthening partnerships with top sellers while developing mid-tier sellers can reduce revenue concentration risk and improve platform stability.

Freight Cost vs Purchase Value

Freight costs do not scale proportionally with purchase value, indicating inefficiencies in shipping pricing. Free shipping thresholds or optimized freight pricing could help increase AOV without materially increasing logistics cost.

2. Delivery & Logistics Performance

Business Objective: Reduce delays and improve efficiency.

Key Findings

Average Delivery Time

The average delivery time across orders is approximately **12 days**, indicating moderate fulfillment speed with room for improvement.

Late vs On-Time Deliveries

While most orders are delivered on time, a non-trivial share of late deliveries persists. These delays represent operational friction and negatively affect customer experience.

Impact of Late Deliveries on Ratings

Late deliveries receive significantly lower customer ratings than on-time orders. Delivery speed is a critical determinant of customer satisfaction and directly influences perceived platform quality.

Seller-Driven Delays

A small number of sellers account for a disproportionate share of late deliveries, with late rates exceeding **50–80%**. These sellers pose a reputational and operational risk to the platform.

Delivery Performance Over Time

Although delivery times improved substantially over the years (from ~19 days to ~11 days), delivery reliability remains a key challenge. Persistent late deliveries silently erode brand trust and increase churn risk.

3. Customer Behavior & Loyalty

Business Objective: Retain customers and increase repeat purchases.

Key Findings

Repeat Customer Rate

Only **13.1%** of customers are repeat buyers, revealing a significant retention gap. Growth is constrained more by customer experience than by customer acquisition.

Repeat Customers by State

Repeat purchase behavior varies by state. Smaller states such as AP show higher repeat rates (~34%), while high-volume states like SP and RS show relatively lower loyalty (~28%). Improving experience in major states offers the largest upside.

Spending Behavior: Repeat vs New Customers

Repeat customers spend nearly **double** per order compared to new customers (~237 vs 129). Despite their high value, repeat customers are vulnerable to churn driven by delivery failures.

Order Frequency

Average order frequency is close to 1, confirming that most customers make only a single purchase. This highlights a “leaky bucket” problem where acquisition outpaces retention.

Tolerance for Delivery Delays

Repeat customers are **not more tolerant** of delivery delays. In fact, they penalize late deliveries with lower ratings, indicating higher service expectations and greater lifetime value risk.

Categories Driving Loyalty

Home and security-related categories show stronger repeat purchase behavior. These categories present opportunities for retention-focused strategies and loyalty programs.

4. Payment & Transaction Behavior

Business Objective: Optimize payment methods.

Key Findings

Payment Method Usage & Revenue

Credit cards dominate both transaction volume and revenue, making them the platform’s most critical payment method. Boleto contributes a smaller but meaningful share, while debit card and voucher usage is minimal.

Installments and Order Value

Installment payments significantly increase AOV. Orders paid in installments average **178.04**, compared to **107.53** for single-payment orders. Installments enable higher-priced purchases and larger baskets.

Payment Type and Cancellations

Order cancellations are extremely rare (6 out of 96,461 orders) and show no meaningful relationship with payment type. Payment method is not a driver of cancellations in this dataset.

Payment Method and Delivery Speed

Late delivery rates vary by payment type. Boleto and voucher payments show higher late delivery percentages compared to card-based payments, suggesting that payment confirmation timing may influence fulfillment speed.

Overall Business Takeaways

- Revenue, sellers, and demand are highly concentrated
- Delivery delays are the strongest driver of customer dissatisfaction
- Customer retention is the largest growth bottleneck

- Repeat customers generate substantially higher value
 - Installment payments increase revenue but require operational discipline
 - Poor-performing sellers create outsized operational and reputational risk
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Conclusion

This project demonstrates an end-to-end business analytics workflow, translating raw e-commerce data into actionable insights across sales, logistics, customers, and payments. The findings indicate that sustainable growth depends less on acquiring new customers and more on improving delivery reliability, enforcing seller performance standards, and protecting high-value repeat customers.